

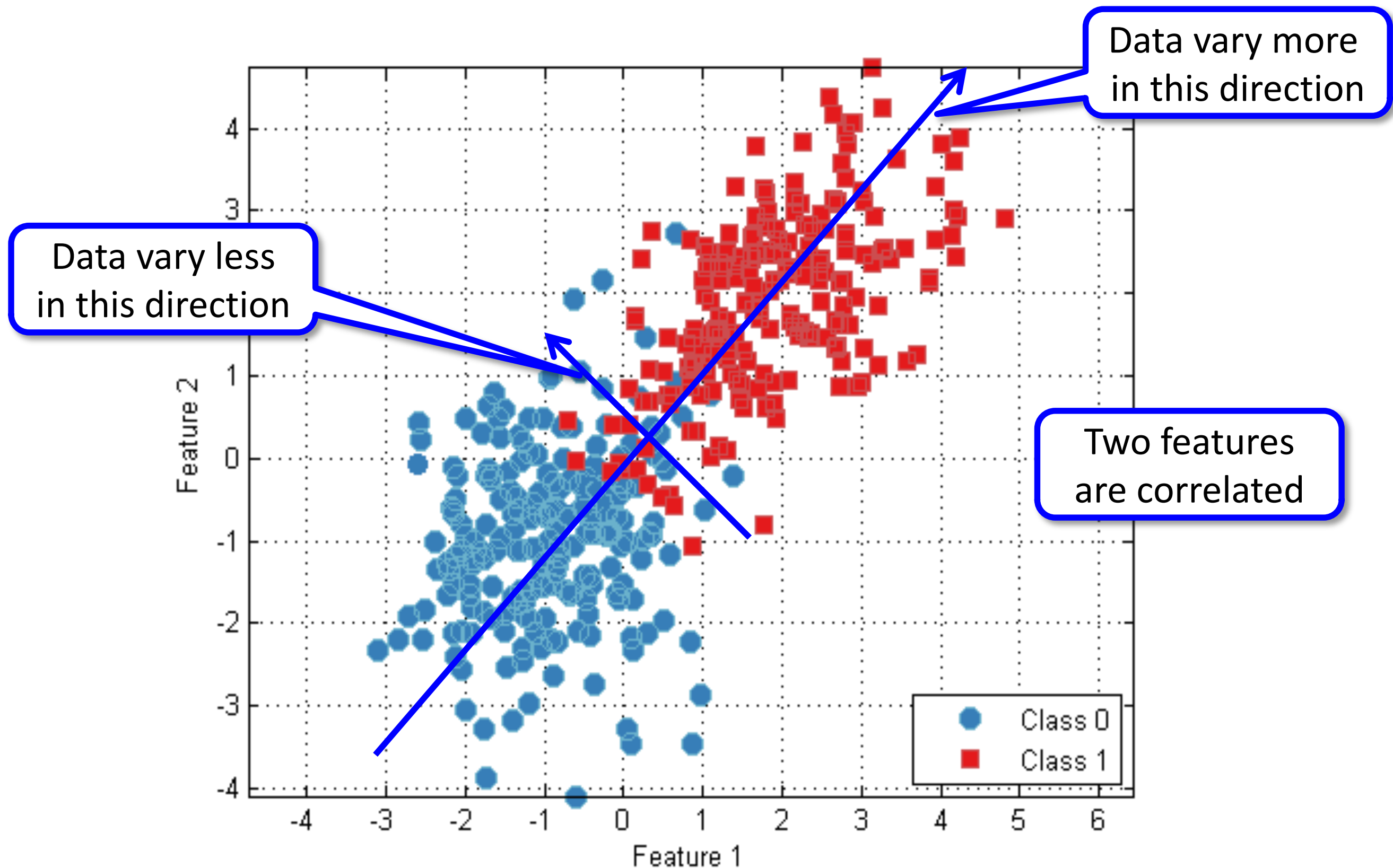
CSE 6740: Computational Data Analysis

Spring 2026

Non-linear Dimensionality Reduction

Anqi Wu
01/29

Principal direction of variation in the dataset



Principal component analysis

- Given m data points, $\{x^1, x^2, \dots, x^m\} \in R^d$, with mean

- Step 1: Estimate the mean and covariance matrix from data

$$\mu = \frac{1}{m} \sum_{i=1}^m x^i \quad \text{and} \quad C = \frac{1}{m} \sum_{i=1}^m (x^i - \mu)(x^i - \mu)^\top$$

Principal directions

- Step 2: Take the eigenvectors $w^1, w^2, \dots$ of C corresponding to the largest eigenvalue λ_1 , the second largest eigenvalue $\lambda_2 \dots$

- Step 3: Compute reduced representation

$$z^i = \begin{pmatrix} w^{1\top} (x^i - \mu) / \sqrt{\lambda_1} \\ w^{2\top} (x^i - \mu) / \sqrt{\lambda_2} \\ \vdots \end{pmatrix}$$

Normalize by
standard deviation

Formulation of PCA

- Given m data points, $\{x^1, x^2, \dots, x^m\} \in R^d$, with their mean $\mu = \frac{1}{m} \sum_{i=1}^m x^i$
- Find a direction $w \in R^d$ where $\|w\| = 1$
- Such that the variance (or variation) of the data along direction w is maximized

$$\max_{w: \|w\|=1} \frac{1}{m} \sum_{i=1}^m (w^\top x^i - w^\top \mu)^2 = \frac{1}{m} w^\top C w$$

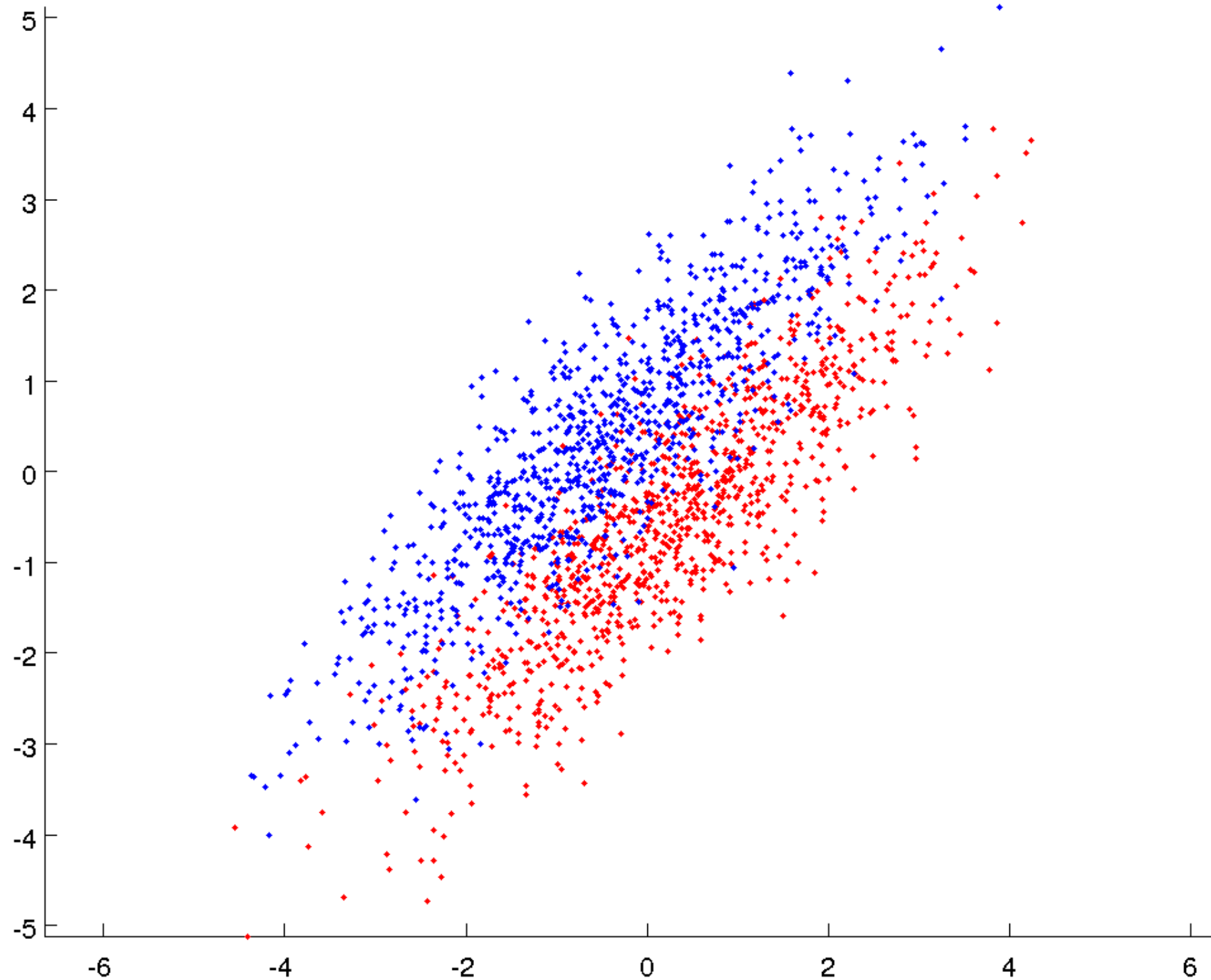
- Solve via a eigenvalue decomposition problem

$$Cw = \lambda w$$

Multiple principal directions

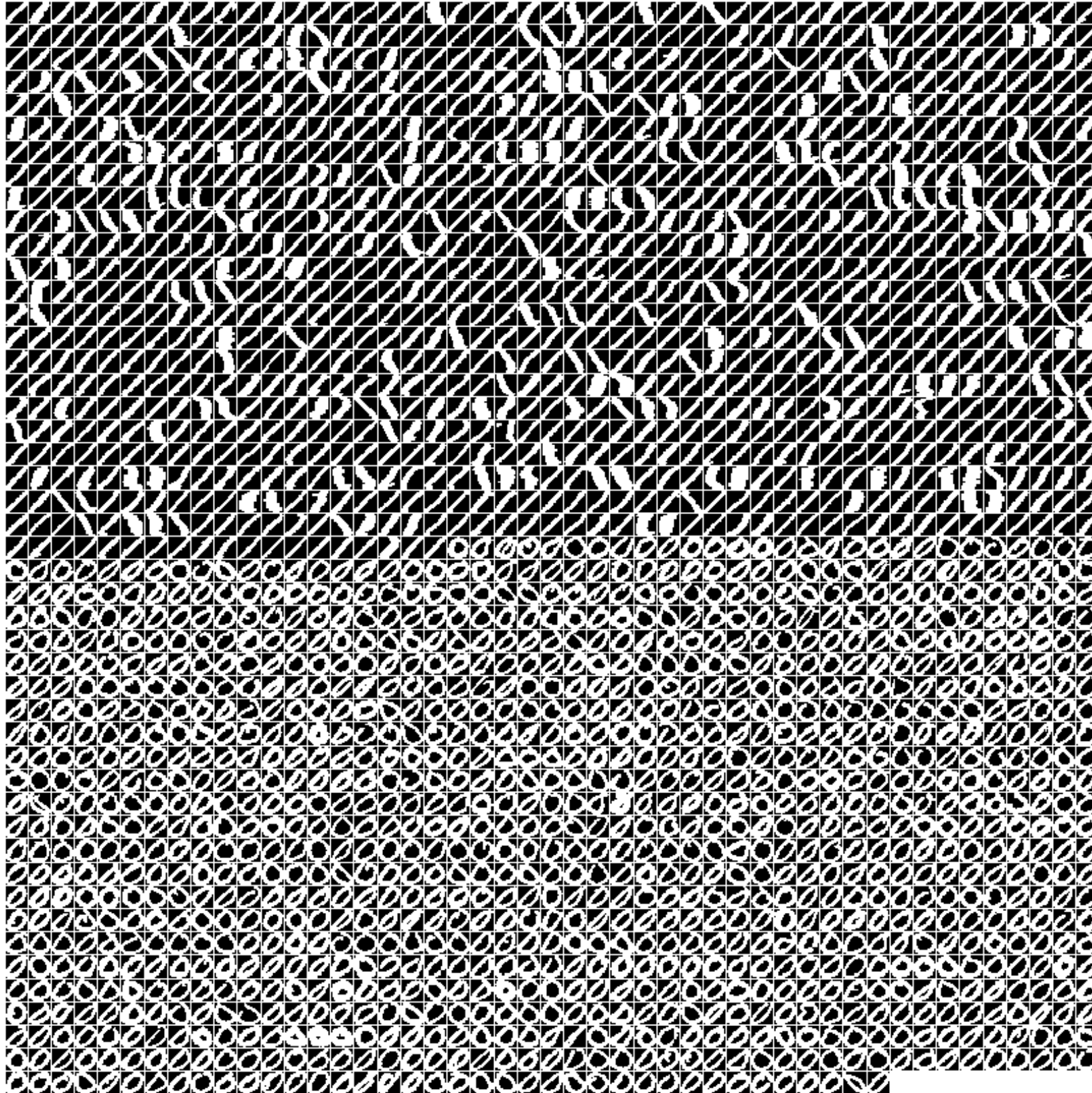
- Directions $w^1, w^2, \dots$ which has
 - the largest variance, the second largest variance, ...
 - but are **orthogonal** to each other
- Take the eigenvectors $w^1, w^2, \dots$ of C corresponding to
 - the largest eigenvalue λ_1 , the second largest eigenvalue λ_2 ...
 - Eigenvalue λ_i = variance of the data in direction w_i

Are principal components good for classification?



Run demo PCA_digits.m

digit 1 and 0





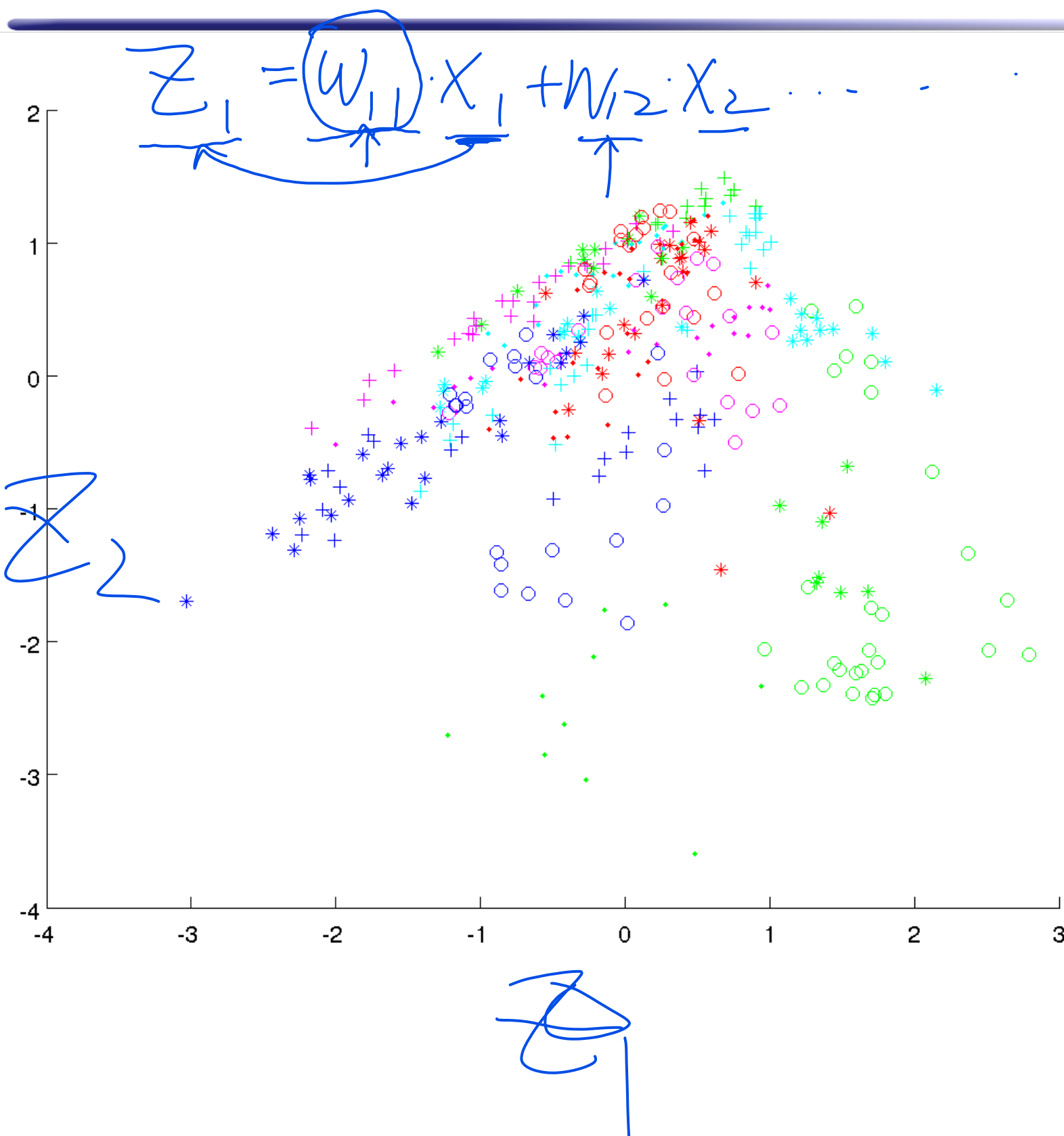
Input features (representation)

Shape feature	Description
<i>Eccentricity</i>	Eccentricity of the ellipse with identical second moments to I . This value ranges from 0 to 1.
<i>Aspect Ratio</i>	Consider any $X, Y \in \partial I$. Choose X and Y such that $d(X, Y) = D(I)$. Find $Z, W \in \partial I$ maximizing $D^\perp = d(Z, W)$ on the set of all pairs of ∂I that define a segment orthogonal to $[XY]$. The aspect ratio is defined as the quotient $D(I)/D^\perp$. Values close to 0 indicate an elongated shape.
<i>Elongation</i>	Compute the maximum escape distance $d_{\max} = \max_{X \in I} d(X, \partial I)$. Elongation is obtained as $1 - 2d_{\max}/D(I)$ and ranges from 0 to 1. The minimum is achieved for a circular region. Note that the ratio $2d_{\max}/D(I)$ is the quotient between the diameter of the largest inscribed circle and the diameter of the smallest circumscribed circle.
<i>Solidity</i>	The ratio $A(I)/A(H(I))$ is computed, which can be understood as a certain measure of convexity. It measures how well I fits a convex shape.
<i>Stochastic Convexity</i>	This variable extends the usual notion of convexity in topological sense, using sampling to perform the calculation. The aim is to estimate the probability of a random segment $[XY]$, $X, Y \in I$, to be fully contained in I .
<i>Isoperimetric Factor</i>	The ratio $4\pi A(I)/L(\partial I)^2$ is calculated. The maximum value of 1 is reached for a circular region. Curvy intertwined contours yield low values.
<i>Maximal Indentation Depth</i>	Let $C_{H(I)}$ and $L(H(I))$ denote the centroid and arclength of $H(I)$. The distances $d(X, C_{H(I)})$ and $d(Y, C_{H(I)})$ are computed $\forall X \in H(I)$ and $\forall Y \in \partial I$. The indentation function can then be defined as $[d(X, C_{H(I)}) - d(Y, C_{H(I)})]/L(H(I))$, which is sampled at one degree intervals. The maximal indentation depth $\mathfrak{D}$ is the maximum of this function.
<i>Lobedness</i>	The Fourier Transform of the indentation function above is computed after mean removal. The resulting spectrum is normalized by the total energy. Calculate lobedness as $F \times \mathfrak{D}^2$, where F stands for the smallest frequency at which the cumulated energy exceeds 80%. This feature characterizes how lobed a leaf is.

Texture feature	Description
<i>Average Intensity</i>	Average intensity is defined as the mean of the intensity in the region.
<i>Average Contrast</i>	Average contrast is the the standard deviation of the intensity in the region, $\sigma = \sqrt{\mu_2(z)}$.
<i>Smoothness</i>	Smoothness is defined as $R = 1 - 1/(1 + \sigma^2)$ and measures the relative smoothness of the intensities in a given region. For regions of constant intensity, R takes the value 0 and R approaches 1 as the intensity values in the region become more uniform. σ^2 is normalized by $(L - 1)^2$ to ensure that $R \in [0, 1]$.
<i>Third moment</i>	μ_3 is a measure of the intensity histogram's skewness. This is generally normalized by $(L - 1)^2$ like smoothness.
<i>Uniformity</i>	Defined as $U = \sum_{i=0}^{L-1} p^2(z_i)$, uniformity's maximum is reached when all intensity levels are equal.
<i>Entropy</i>	A measure of intensity randomness.

8 shape features and
6 texture features

Reduce representation



Principal direction:

$W =$

W_1	W_2
0.0938	0.1924
0.1902	0.0253
0.2266	-0.1800
-0.1850	0.4084
-0.1600	0.3825
-0.2063	0.3488
0.1940	-0.4037
0.2150	-0.3566
-0.3723	-0.2001
-0.3657	-0.1974
-0.3602	-0.2037
-0.3175	-0.1886
-0.3056	-0.1243
-0.3482	-0.1829

Shape
features

Texture
features

Singular Value Decomposition

- Singular value decomposition, known as SVD, is a factorization of a real matrix with applications in calculating pseudo-inverse, rank, solving linear equations, and many others.
- For a matrix $M \in \mathbb{R}^{m \times n}$ assume $n \leq m$
 - $M = U\Sigma V^T$ where $U \in \mathbb{R}^{m \times m}$, $V^T \in \mathbb{R}^{n \times n}$, $\Sigma \in \mathbb{R}^{m \times n}$
 - The m columns of U , and the n columns of V are called the left and right singular vectors of M . The diagonal elements of Σ , Σ_{ii} are known as the singular values of M .
 - Let v be the i^{th} column of V , and u be the i^{th} column of U , and σ be the i^{th} diagonal element of Σ
$$Mv = \sigma u \quad \text{and} \quad M^T u = \sigma v$$

Singular Value Decomposition - II

$$M = \underbrace{[u_1 \ u_2 \ \dots \ u_m]}_{\text{principal directions}} \underbrace{\begin{bmatrix} \Sigma_{11} & \dots & \Sigma_{1n} \\ \vdots & \ddots & \vdots \\ \Sigma_{m1} & \dots & \Sigma_{mn} \end{bmatrix}}_{\text{Scaling factor}} \underbrace{[v_1 \ v_2 \ \dots \ v_n]^T}_{\text{Projection in principal directions}}$$

- Singular value decomposition is related to eigenvalue decomposition

- Suppose $M = [x_1 - \mu \ x_2 - \mu \ \dots \ x_m - \mu] \in \mathbb{R}^{m \times n}$
- Then covariance matrix is $C = \frac{1}{m} M M^T$
- Starting from singular vector pair
 - $M^T u = \sigma v$
 - $\Rightarrow M M^T u = \sigma M v$
 - $\Rightarrow M M^T u = \sigma^2 u$
 - $\Rightarrow C u = \lambda u$

How to recover the original data point?

- Given data mean μ , principal directions $w^1, w^2, \dots$ and the corresponding eigenvalues $\lambda_1, \lambda_2, \dots$
- Recover x^i from the reduced representation z^i **approximately**?

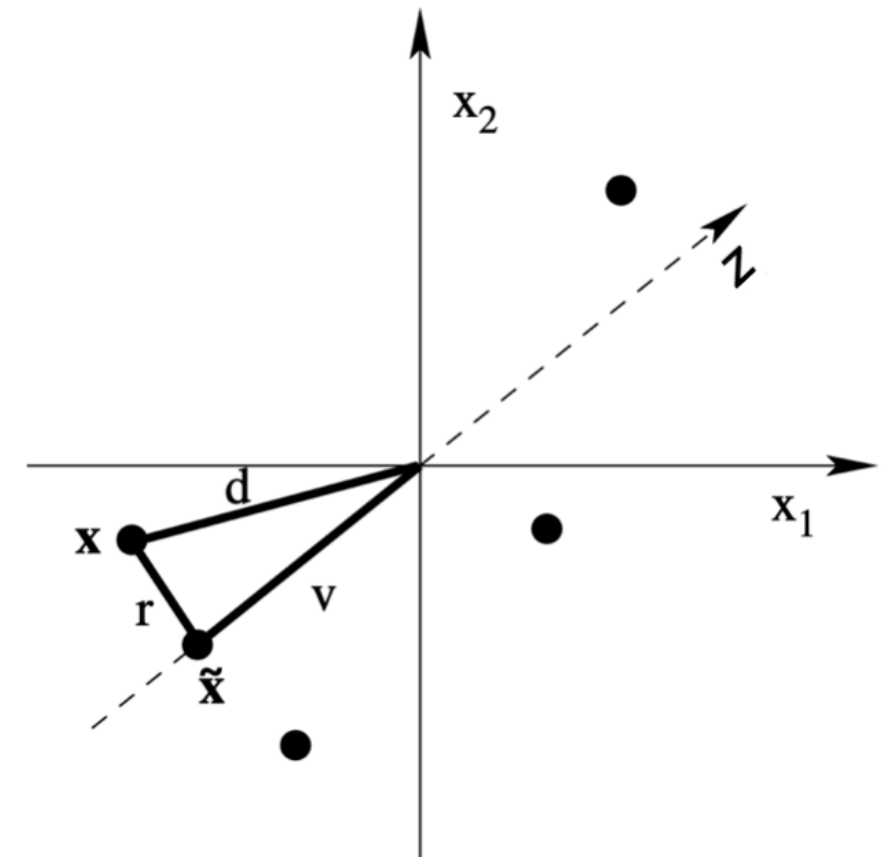
$$z^i = \begin{pmatrix} z_1^i \\ z_2^i \\ \vdots \end{pmatrix} = \begin{pmatrix} w^{1\top} (x^i - \mu) / \sqrt{\lambda_1} \\ w^{2\top} (x^i - \mu) / \sqrt{\lambda_2} \\ \vdots \end{pmatrix}$$

- $x^i \approx \hat{x}^i = \mu + z_1^i \cdot \sqrt{\lambda_1} \cdot w^1 + z_2^i \cdot \sqrt{\lambda_2} \cdot w^2 + \dots$
- Matrix-vector expression?

Two alternatives

- Maximize the variance of the code vectors

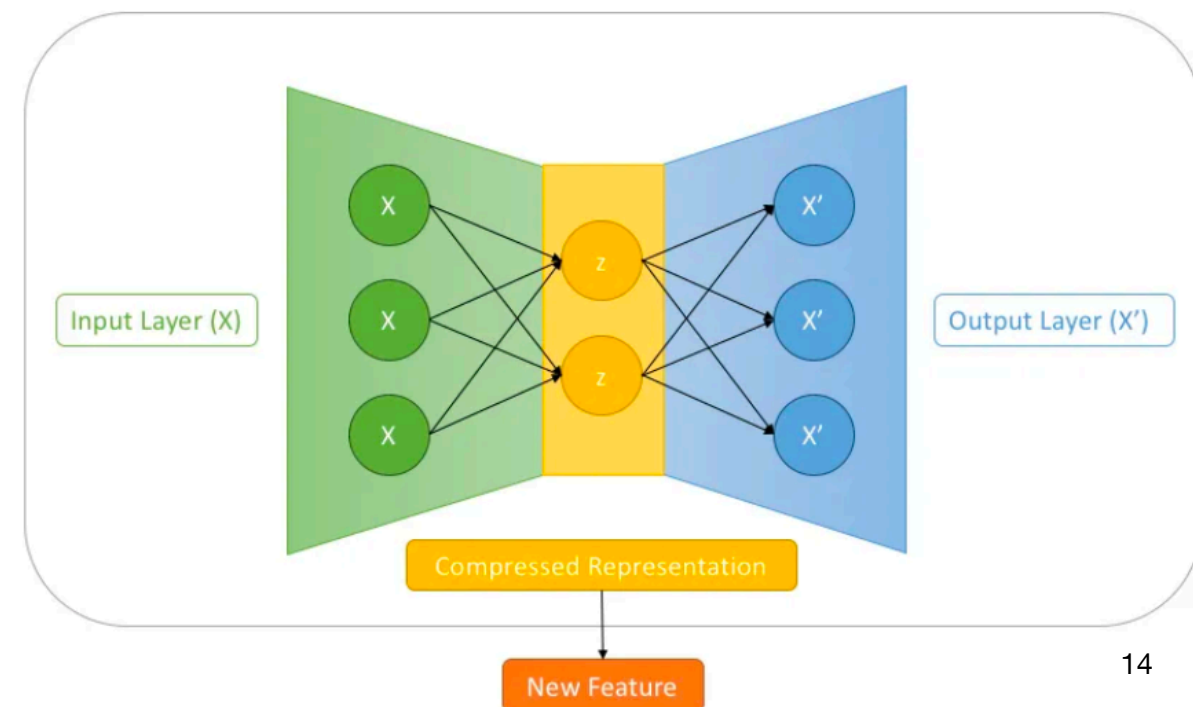
$$\begin{aligned} \max \sum_j \text{Var}(z_j) &= \frac{1}{N} \sum_j \sum_i (z_j^{(i)} - \bar{z}_j)^2 \\ &= \frac{1}{N} \sum_i \|\mathbf{z}^{(i)} - \bar{\mathbf{z}}\|^2 \\ &= \frac{1}{N} \sum_i \|\mathbf{z}^{(i)}\|^2 \end{aligned}$$



- Minimize the reconstruction error

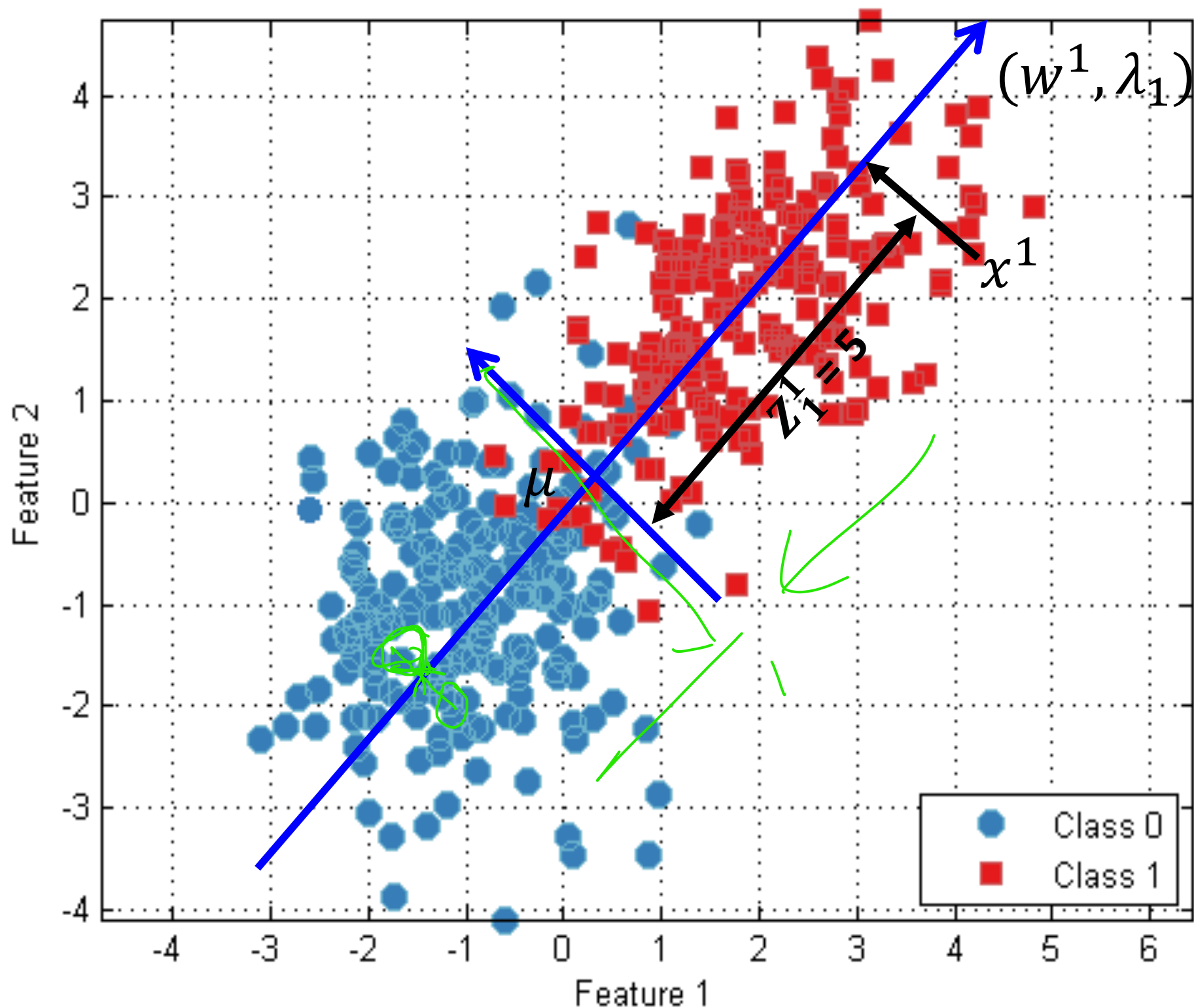
$$\min \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}^{(i)} - \tilde{\mathbf{x}}^{(i)}\|^2$$

Handwritten notes: = d² - v² = d² - v²



Reduce to 1d, reconstruct 2d

$$x^1 \approx \hat{x}^1 = \mu + z_1^1 \cdot \sqrt{\lambda_1} \cdot w^1$$

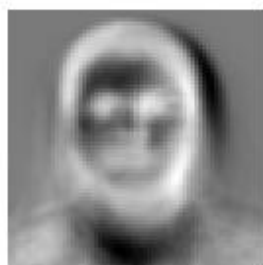


Eigenfaces (demo test_eigenface.m)



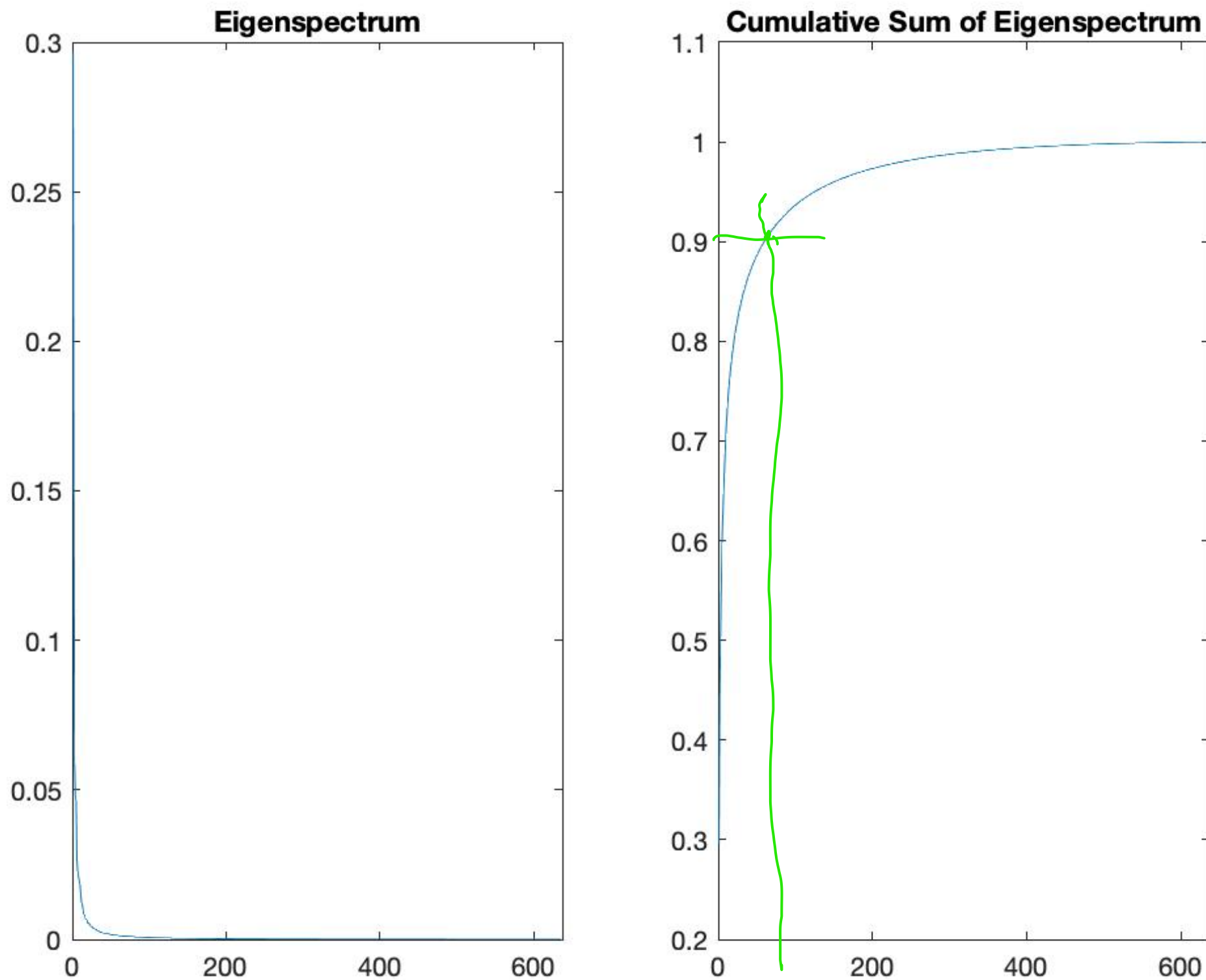
Eigenfaces (demo test_eigenface.m)

Eigenfaces

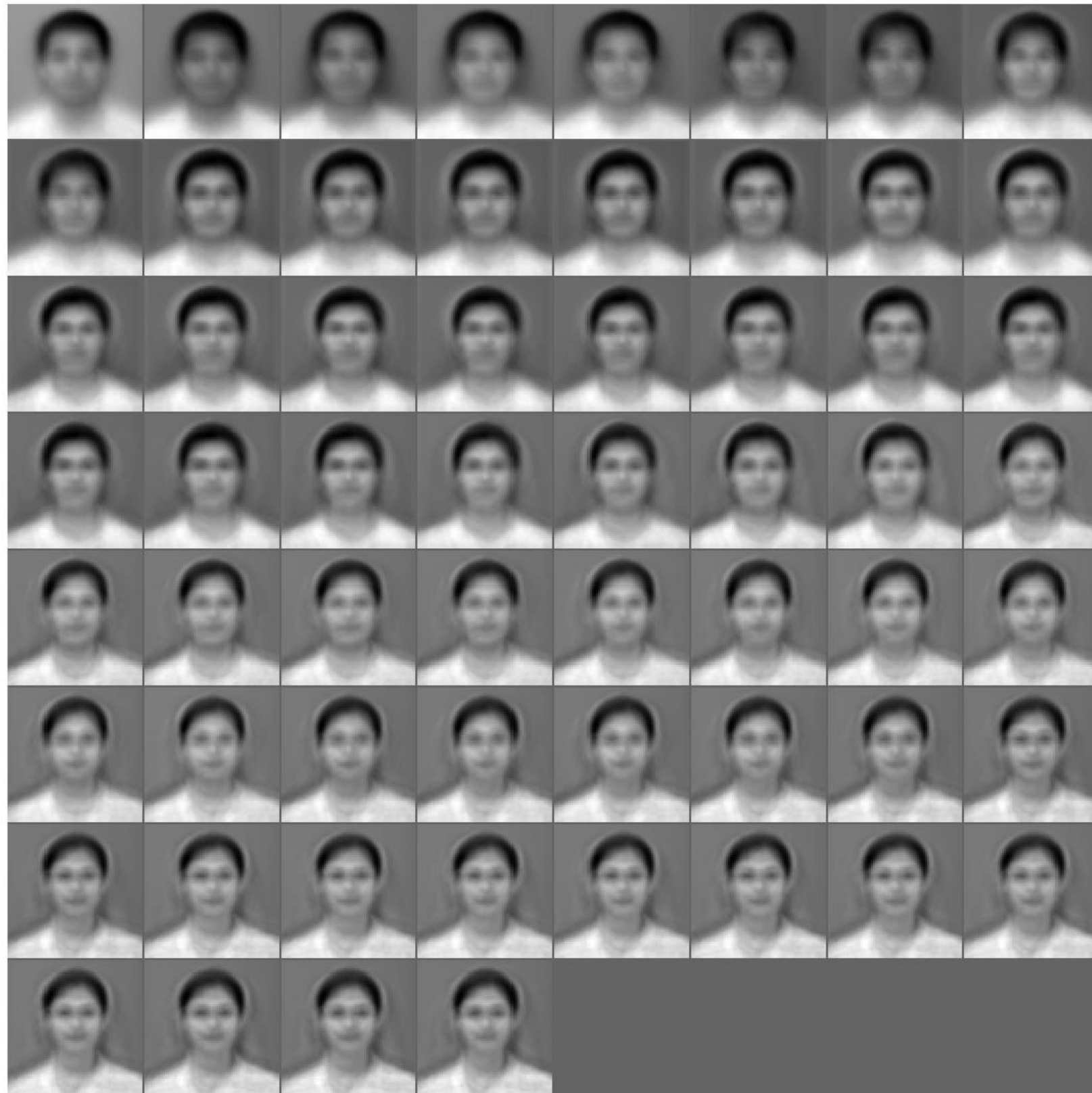


W
b
o

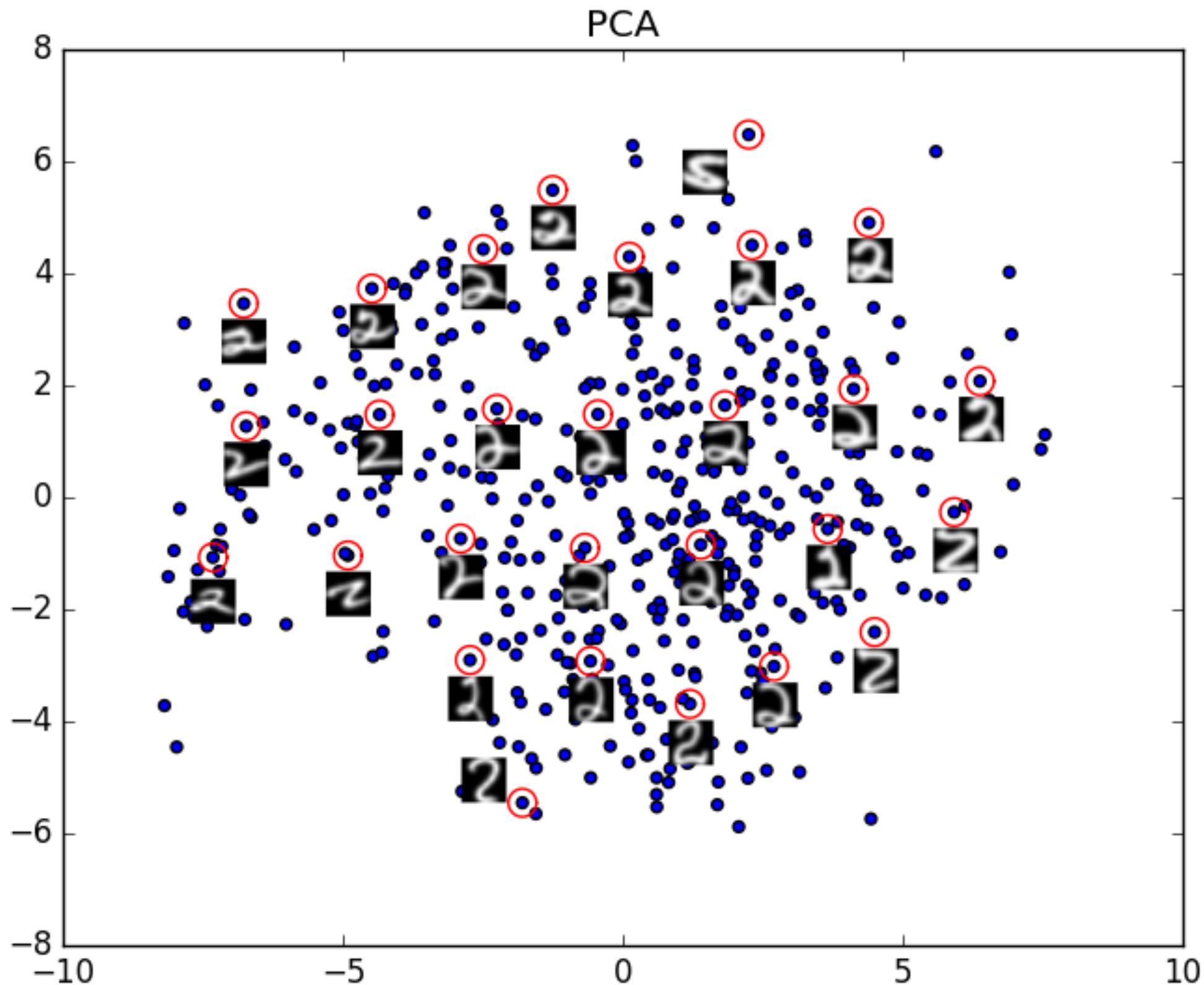
Eigenfaces (demo test_eigenface.m)



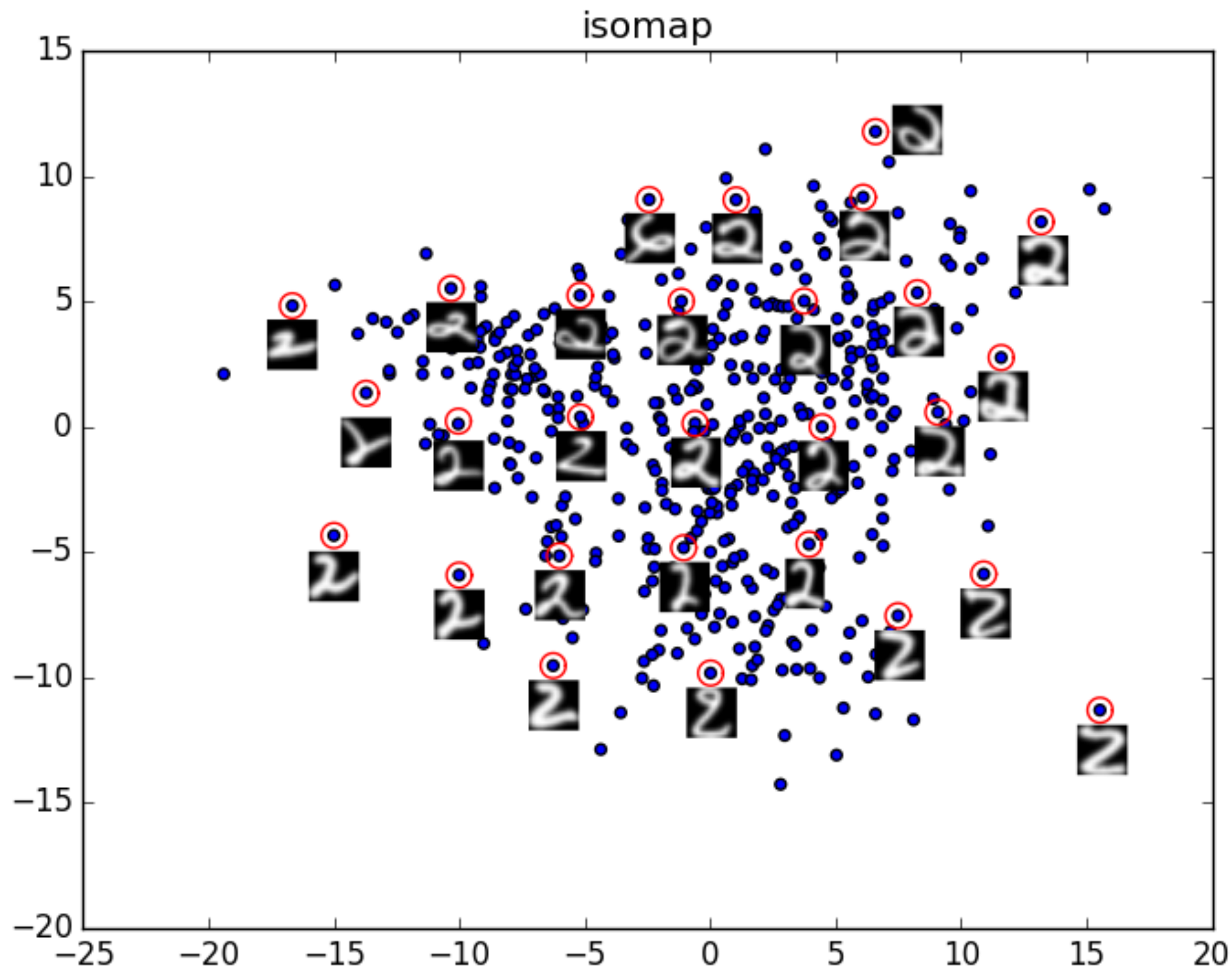
Eigenfaces (demo test_eigenface.m)



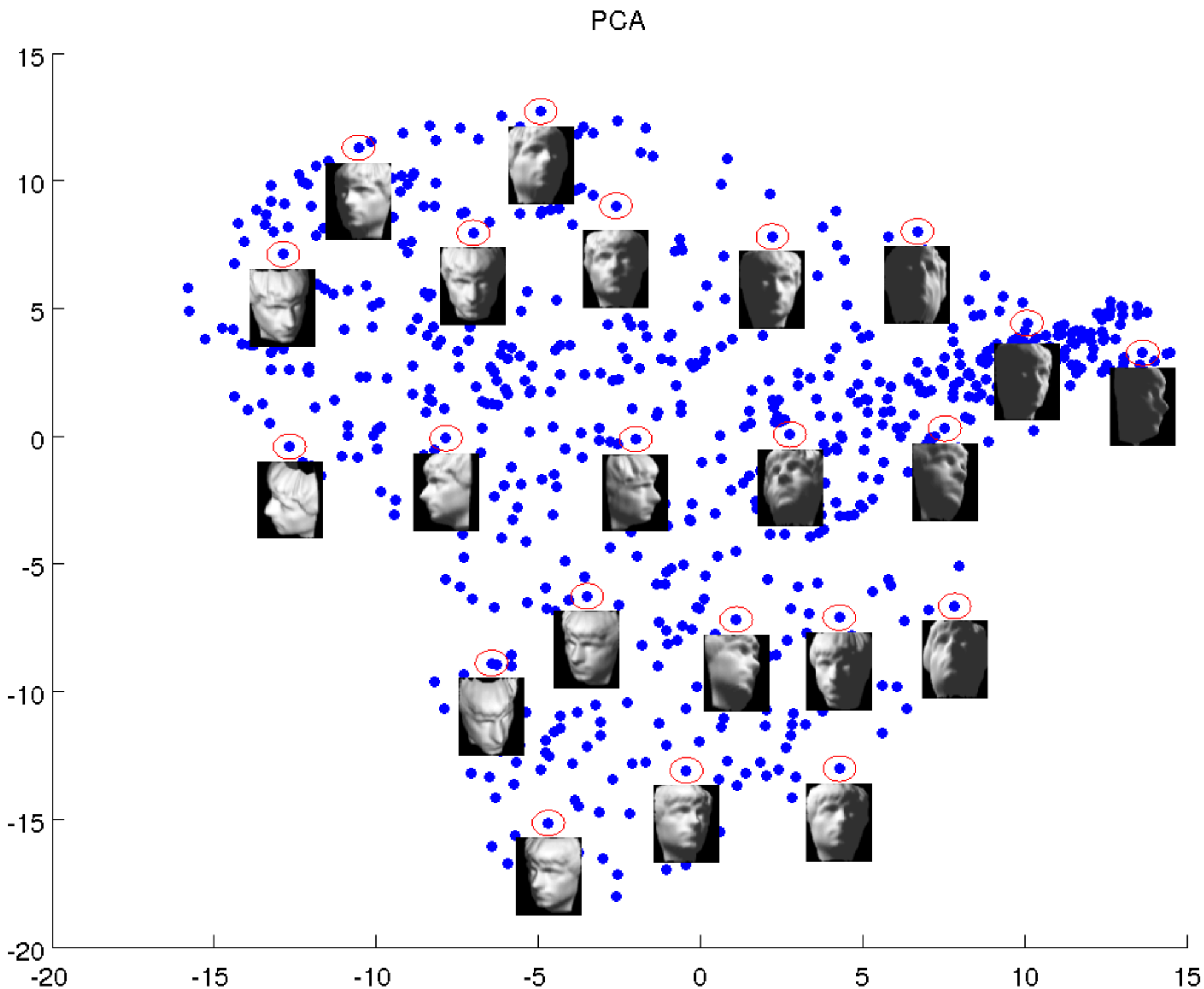
Is the principal direction interpretable?



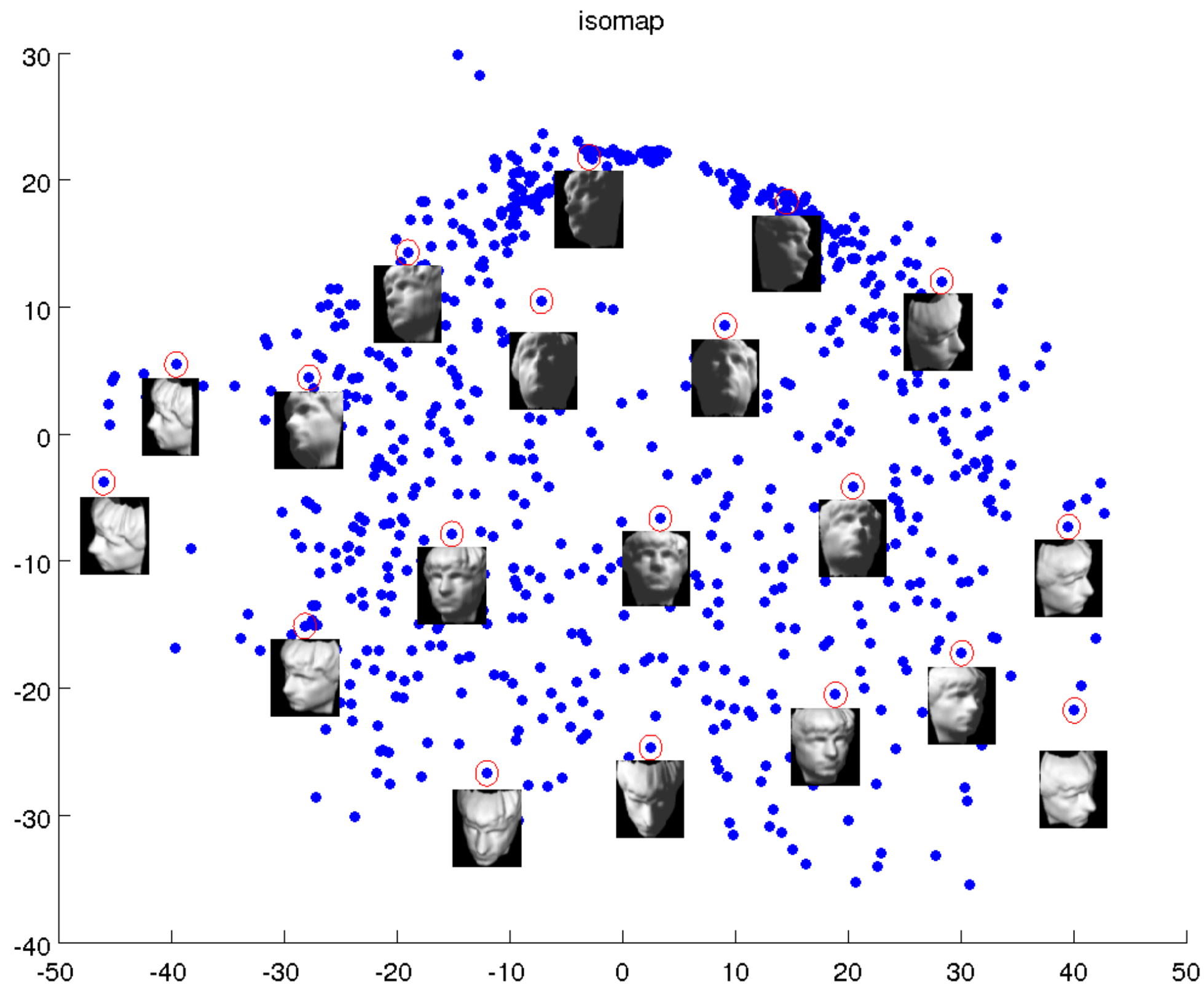
How about this one?



Is the principal direction interpretable?

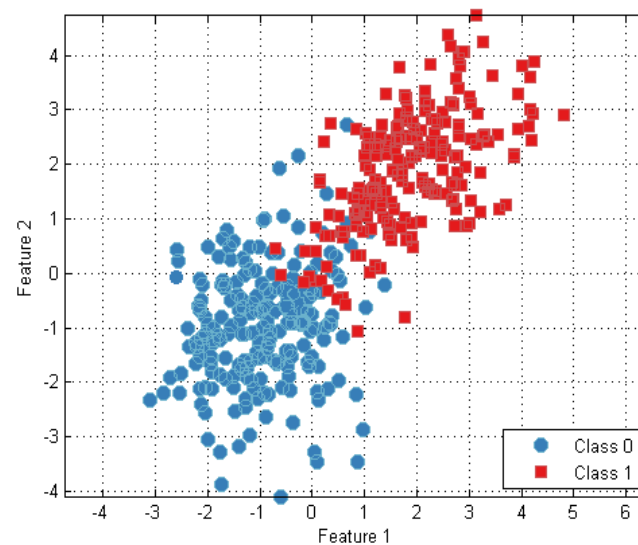


How about this one?

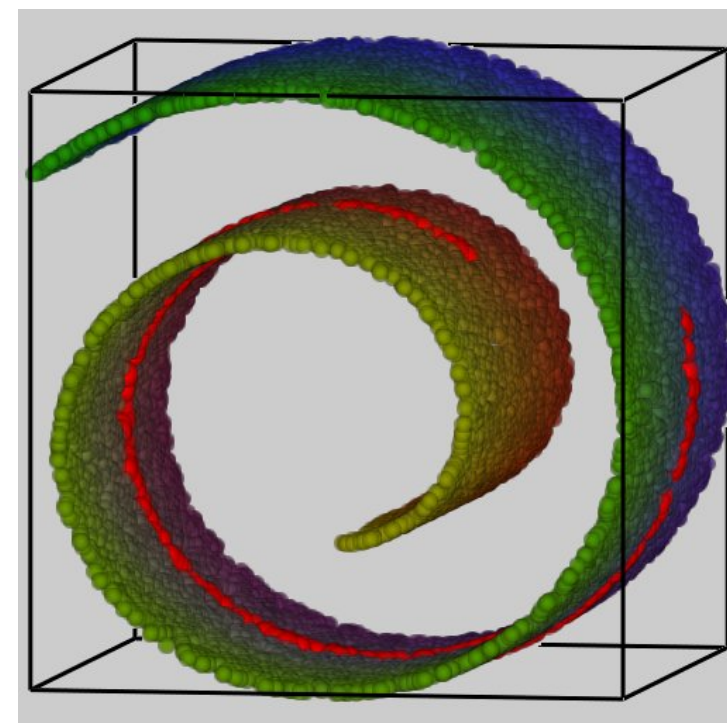
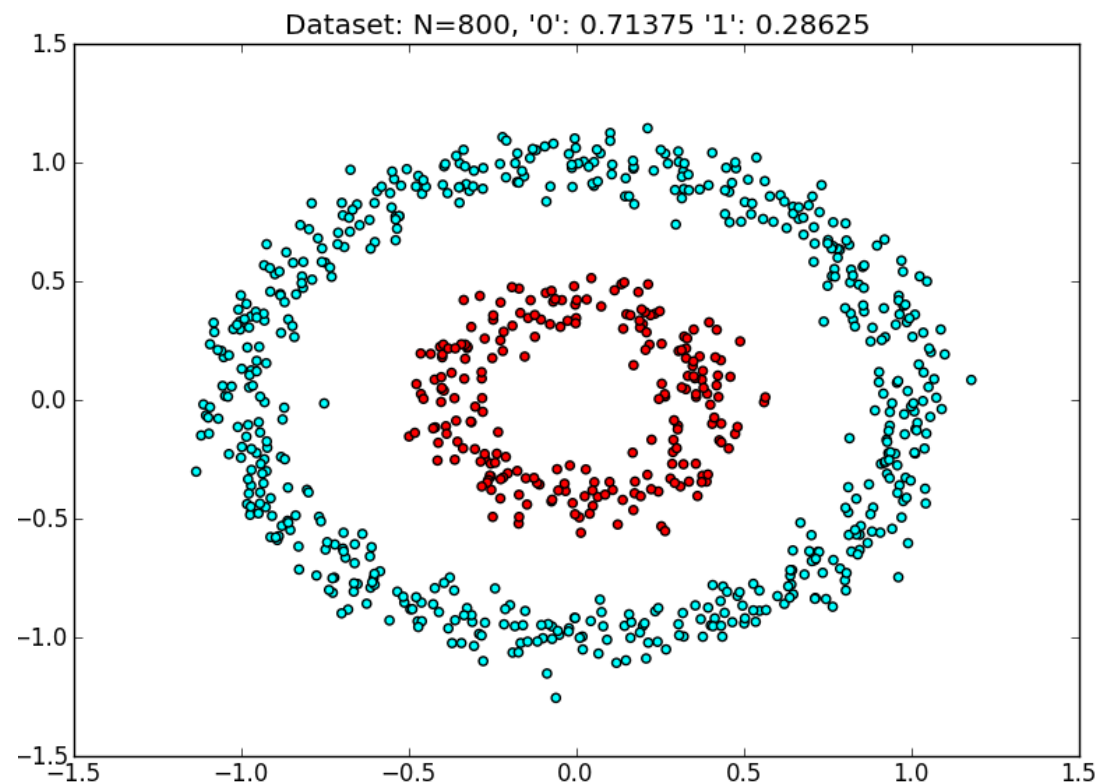


Limitation of PCA and SVD

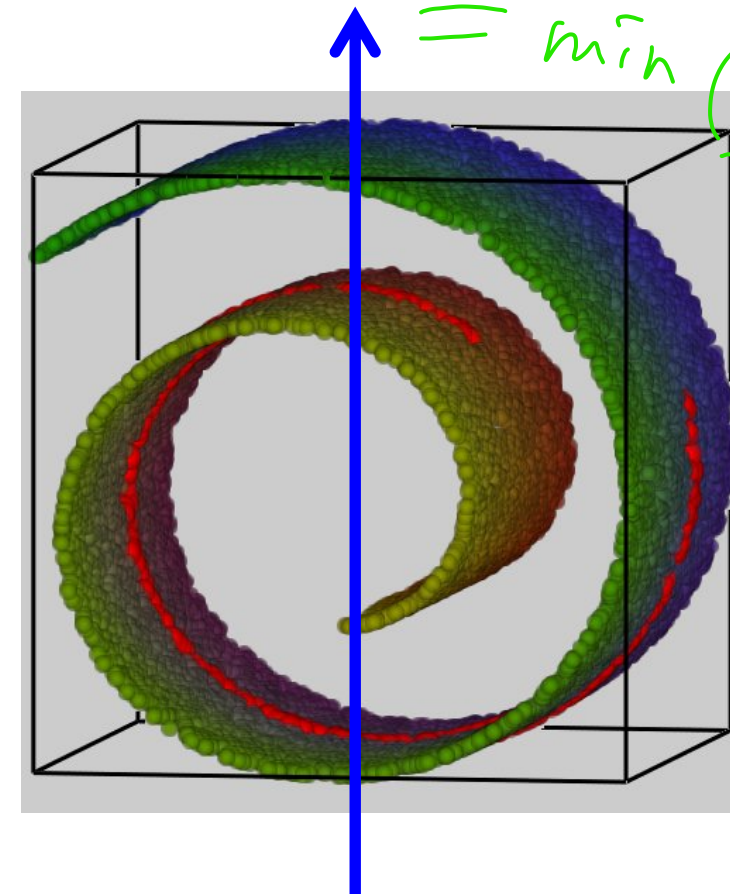
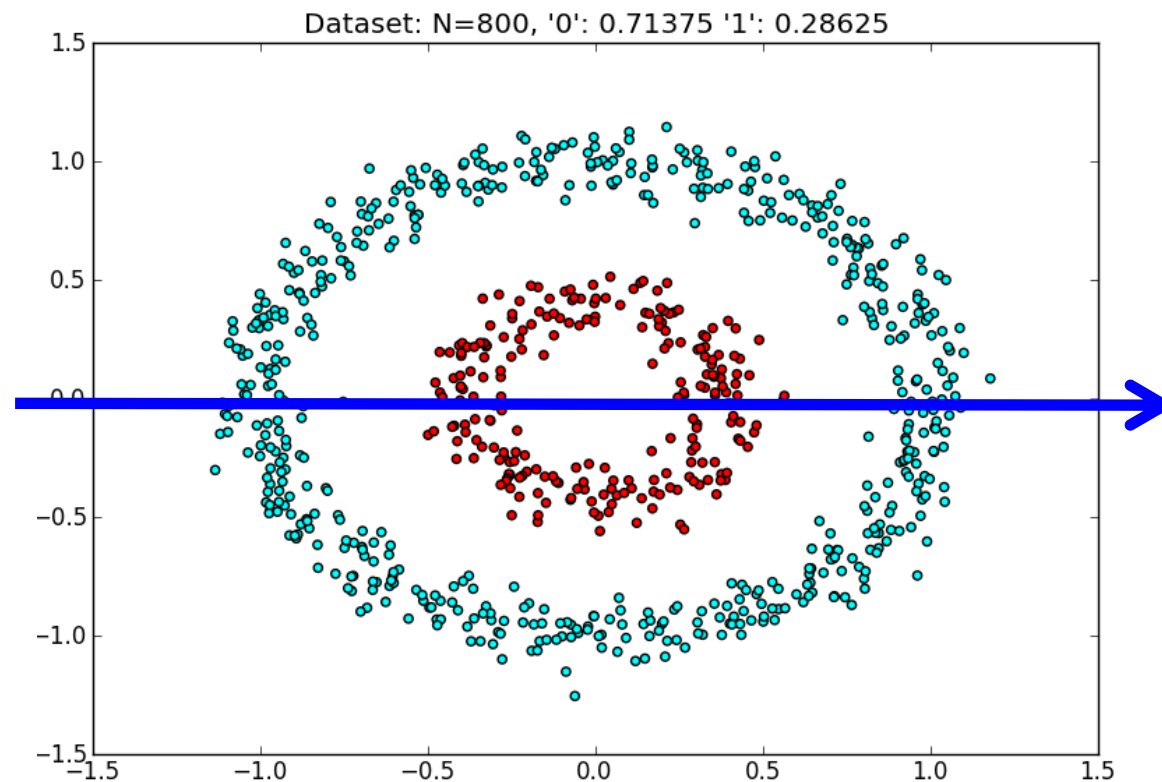
- Suitable when variables are linearly correlated



- Not suitable when nonlinear structures are present

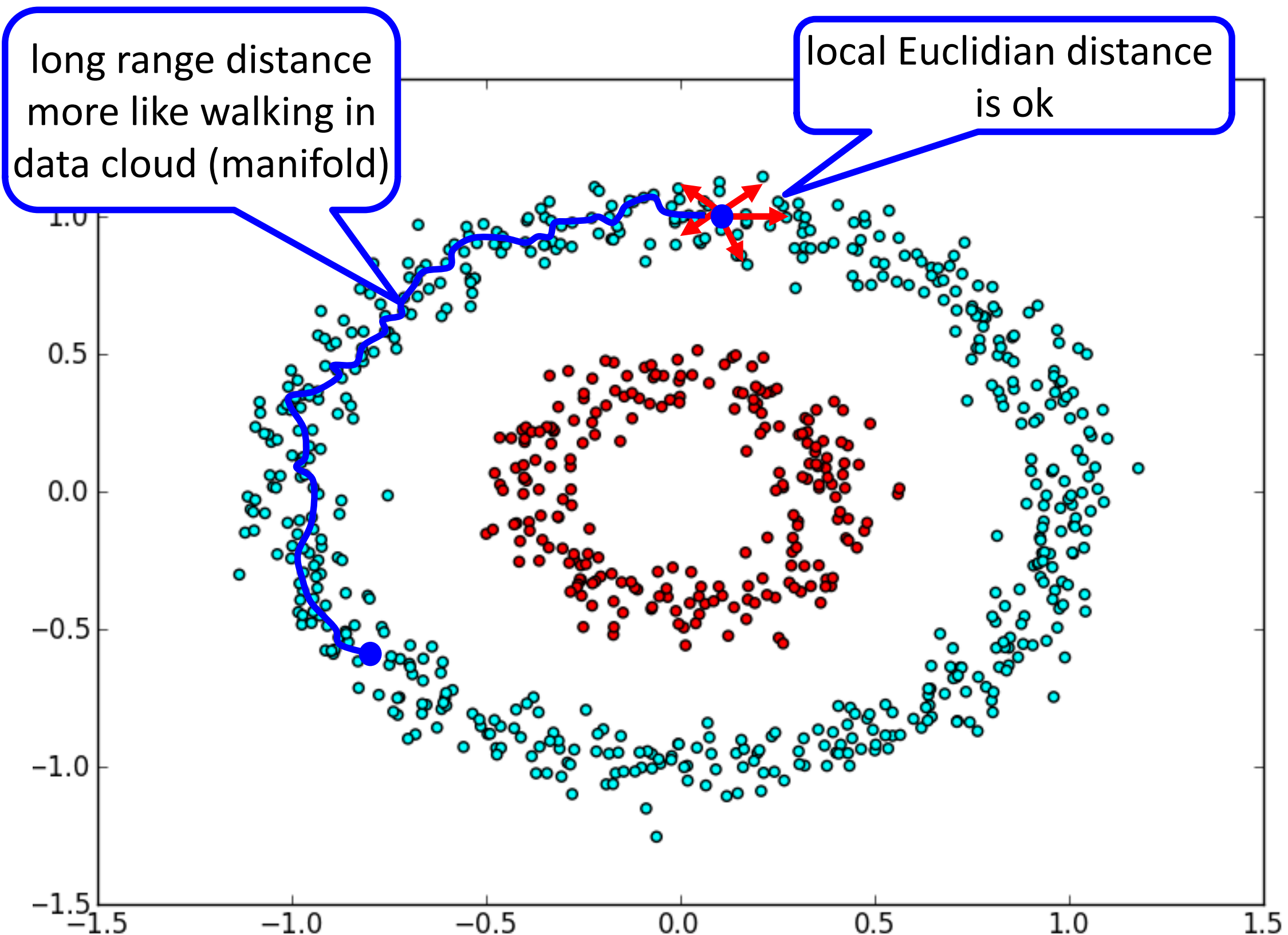


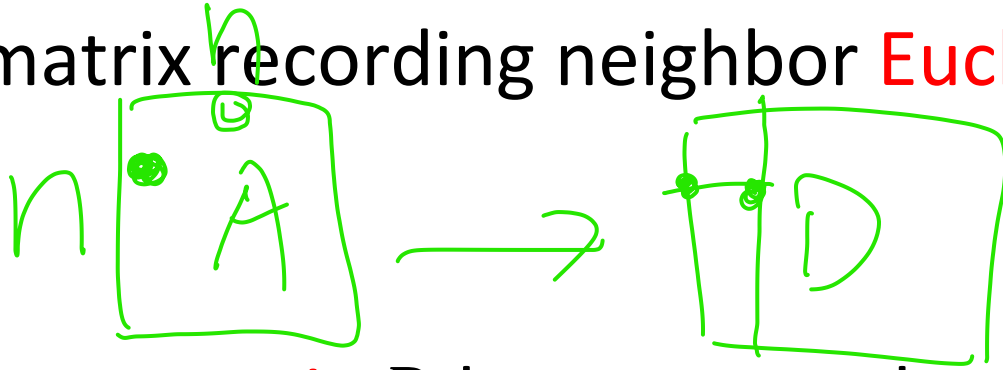
What's wrong with PCA



- PCA uses linear projection $w^T x$, implicitly assuming Euclidean distance is the dissimilarity (distance) measure
- When there are nonlinear structure, Euclidean distance is **not** the right distance measure **globally**

What's a reasonable distance measure

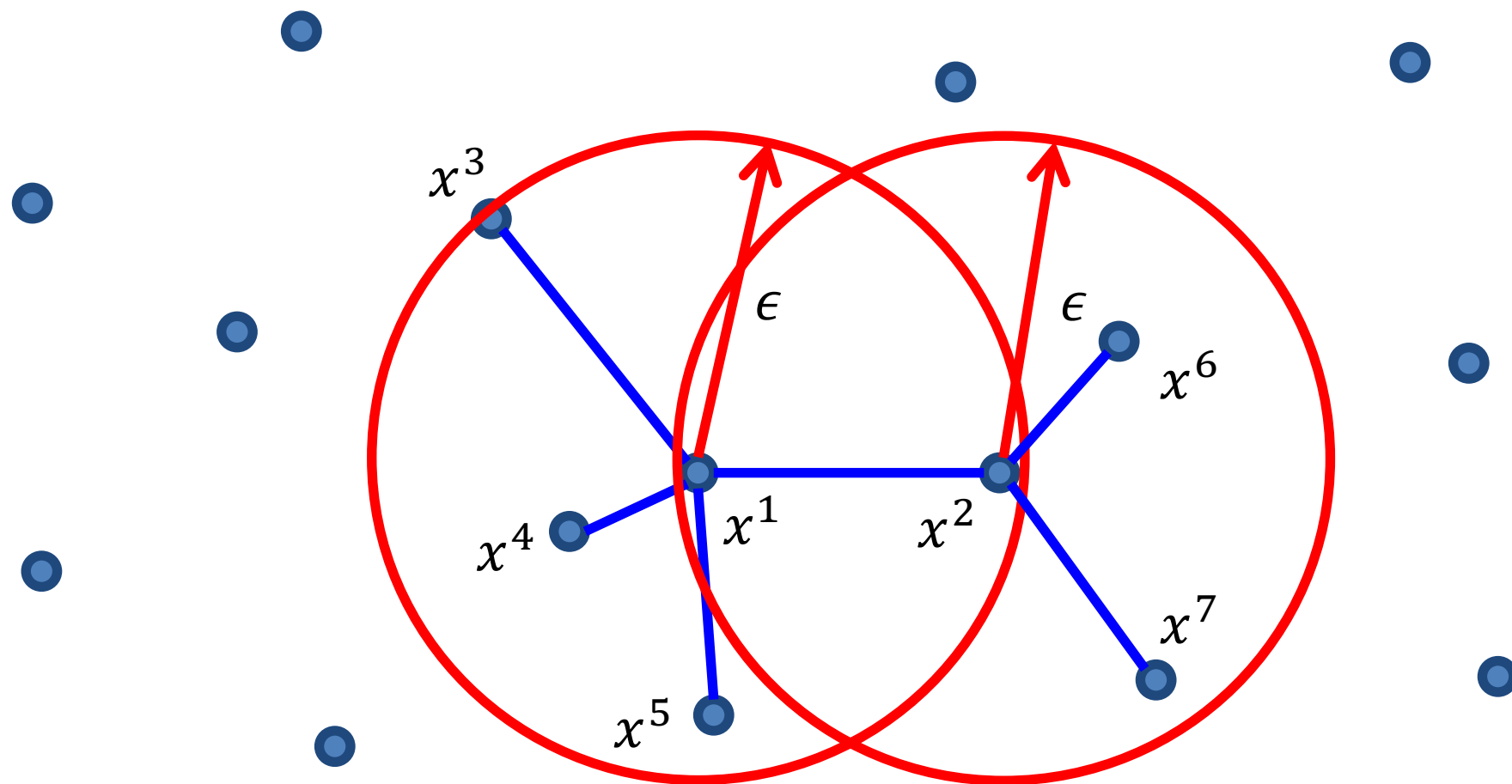


- Key idea: produce low dimensional representation which preserves “walking-distance” over the data cloud (manifold)
- Find neighbors $N(i)$ of each data point, x^i , within distance ϵ and let A be the adjacency matrix recording neighbor **Euclidean distance**

- Find **shortest path distance matrix** D between each pairs of points, x^i and x^j , based on A
- Find low dimensional representation which preserves the distances information in D

Neighbor graph

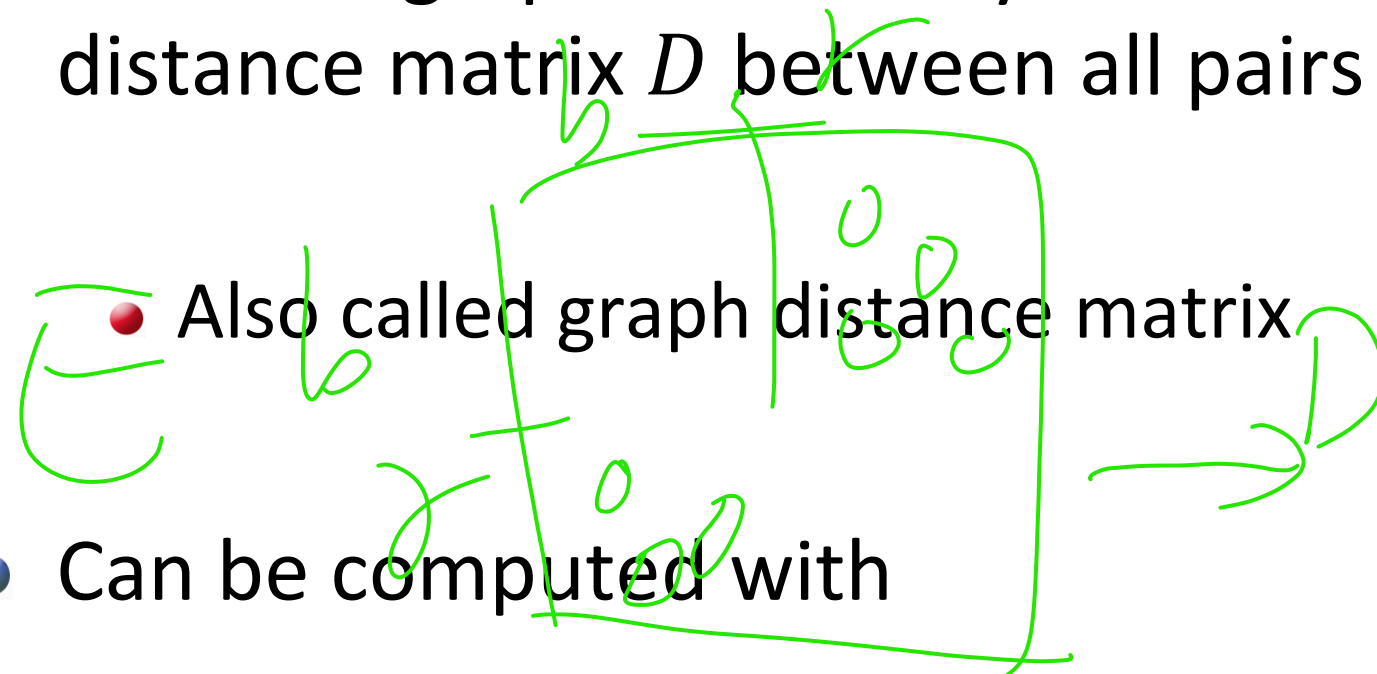
- Given m data points, threshold ϵ , construct matrix $A \in R^{m \times m}$

$$A^{ij} = \begin{cases} \|x^i - x^j\|, & \text{if } \|x^i - x^j\| \leq \epsilon \\ 0, & \text{otherwise} \end{cases}$$



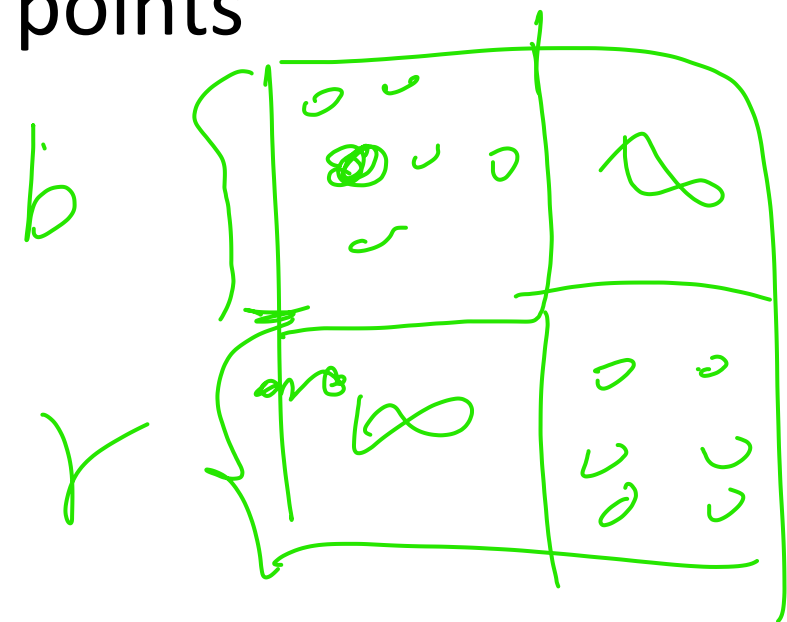
Shortest path distances

- With the graph defined by $A \in \mathbb{R}^{m \times m}$, find the shortest path distance matrix D between all pairs of points



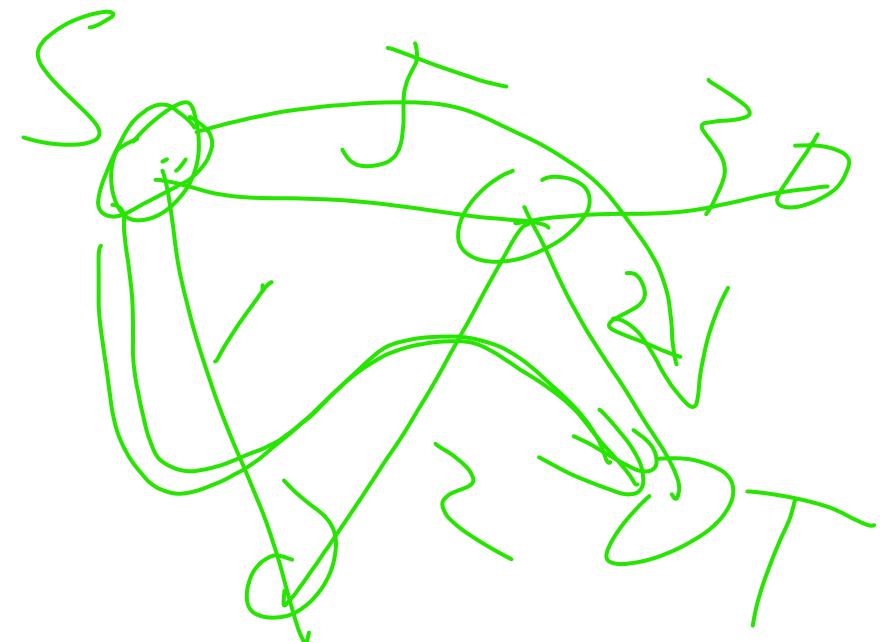
- Also called graph distance matrix

- Can be computed with



- Floyd-Warshall algorithm (all pair shortest path problem)

- m applications of Dijkstra's algorithm



How to extract reduced representation?

- Now we have a distance matrix D
- At this point we can throw away the original data points and work with just D
- Can we extract from D a new set of coordinates (or reduced representation) $\tilde{Z}$ which best explains D ?
- Hints: related distance to inner product.
 - Given coordinates, we can compute Euclidean distances
 - Given Euclidean distances, can we compute the coordinates?

From coordinates to distances

- Distance to inner product

$$\begin{aligned} D^{ij^2} &= \|z^i - z^j\|^2 \\ &= (z^i - z^j)^\top (z^i - z^j) \\ &= z^{i\top} z^i + z^{j\top} z^j - 2z^{i\top} z^j \end{aligned}$$

- In matrix format, let $Z = (z^1, z^2, \dots, z^m) \in \mathbb{R}^{n \times m}$

$$(D)^2 = a1^\top + 1a^\top - 2Z^\top Z$$

Where $a = (z^{1\top} z^1, z^{2\top} z^2, \dots, z^{m\top} z^m)^\top$

Relate distances to coordinates (or inner products)

- Construct a special centering matrix $H = I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top$
- $\left(I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top\right) \mathbf{1} \mathbf{a}^\top \left(I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top\right) = \underline{0}$
- $\left(I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top\right) \mathbf{a} \mathbf{1}^\top \left(I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top\right) = \underline{0}$
- Then apply it to both side of $(D)^2$
- $-\frac{1}{2m} H(D)^2 H = -\frac{1}{2m} H(\mathbf{a} \mathbf{1}^\top + \mathbf{1} \mathbf{a}^\top - 2Z^\top Z) H = \frac{1}{m} H Z^\top Z H$
- $ZH = Z \left(I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top\right) = Z - \mu \mathbf{1}^\top = \tilde{Z}$

Can we uniquely recover the coordinates?

- Given $C = -\frac{1}{2}H(D)^2H = \tilde{Z}^\top \tilde{Z}$
- Multiple coordinate matrices $\tilde{Z}$ result in the same C
- For any orthogonal matrix B :
$$C = \tilde{Z}^\top \tilde{Z} = \tilde{Z}^\top \textcolor{red}{B}^\top \textcolor{red}{B} \tilde{Z}$$
- An orthogonal matrix is like a transformation of the data
- Therefore, we can hope to recover the coordinate matrix $\tilde{Z}$ up to some transformation

Obtain low dimensional representation

- Given $C = -\frac{1}{2}H(D)^2H = \tilde{Z}^T \tilde{Z}$ $\xrightarrow{\text{green}} Z = W \Sigma$
- Pick a **particular** coordinate matrix $\tilde{Z}$ that best approximates C $C = \underline{W} \Lambda W^T$
- Perform eigenvalue decomposition of C
 - $Cw = \lambda w$
 - Take the eigenvectors $w^1, w^2, \dots$ of C corresponding to
 - the largest eigenvalue λ_1 , as the first coordinate
 - the second largest eigenvalue λ_2 , as the second coordinate

- Reduced representation

$$\tilde{Z}^T = (w^1, w^2, \dots) \begin{pmatrix} \lambda_1^{1/2} & & \\ & \lambda_2^{1/2} & \\ & & \ddots \end{pmatrix}$$

Squared Euclidean Distances Yield a PSD Matrix

Let $D^2 \in \mathbb{R}^{m \times m}$ be a squared distance matrix with

$$D_{ij}^2 = \|z_i - z_j\|_2^2,$$

and define the centering matrix

$$H = I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top.$$

Key result:

$$C := -\frac{1}{2} H D^2 H \succeq 0.$$

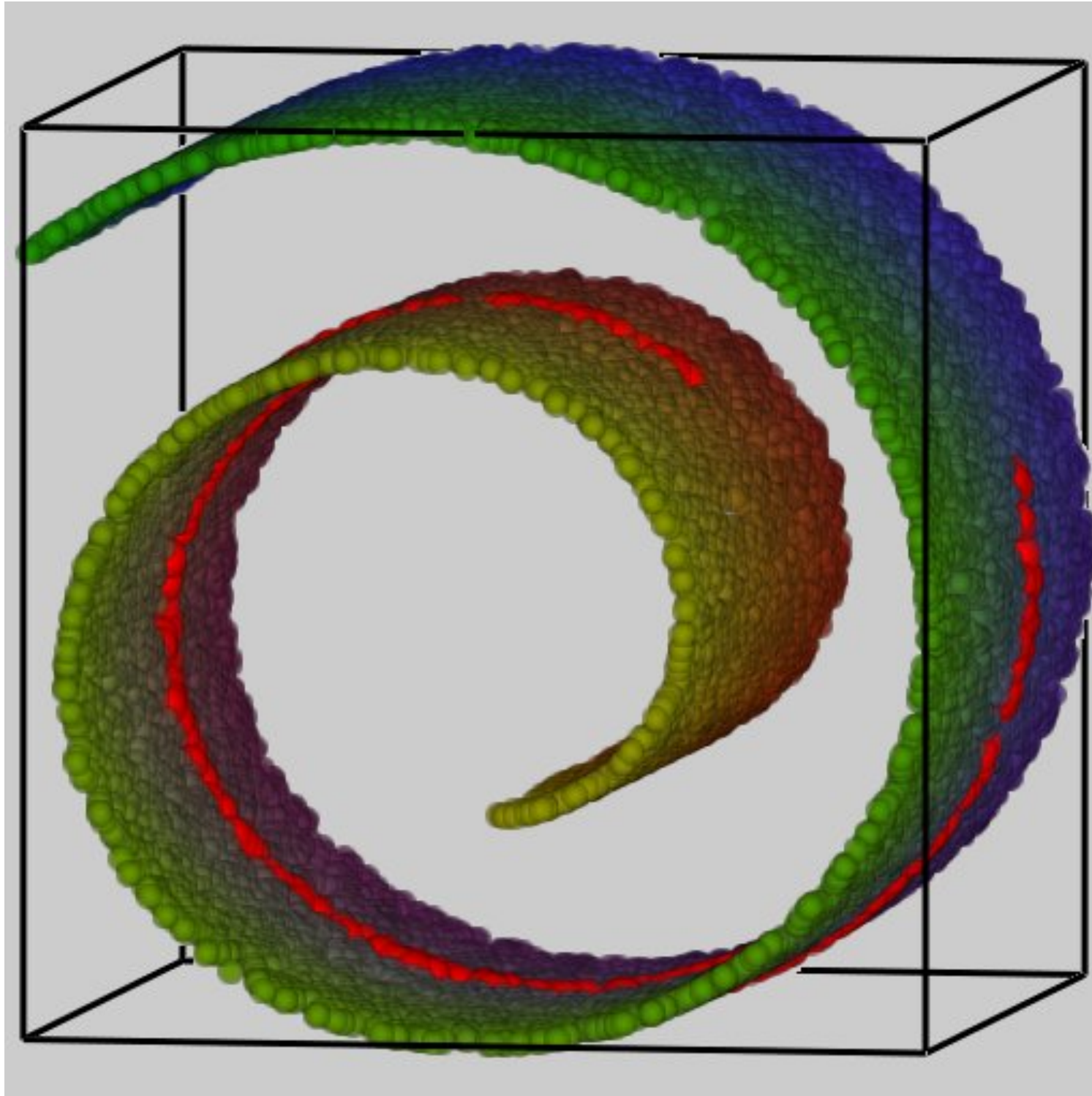
Implication:

- C is positive semidefinite
- All eigenvalues of C are nonnegative
- Coordinates can be recovered via eigendecomposition

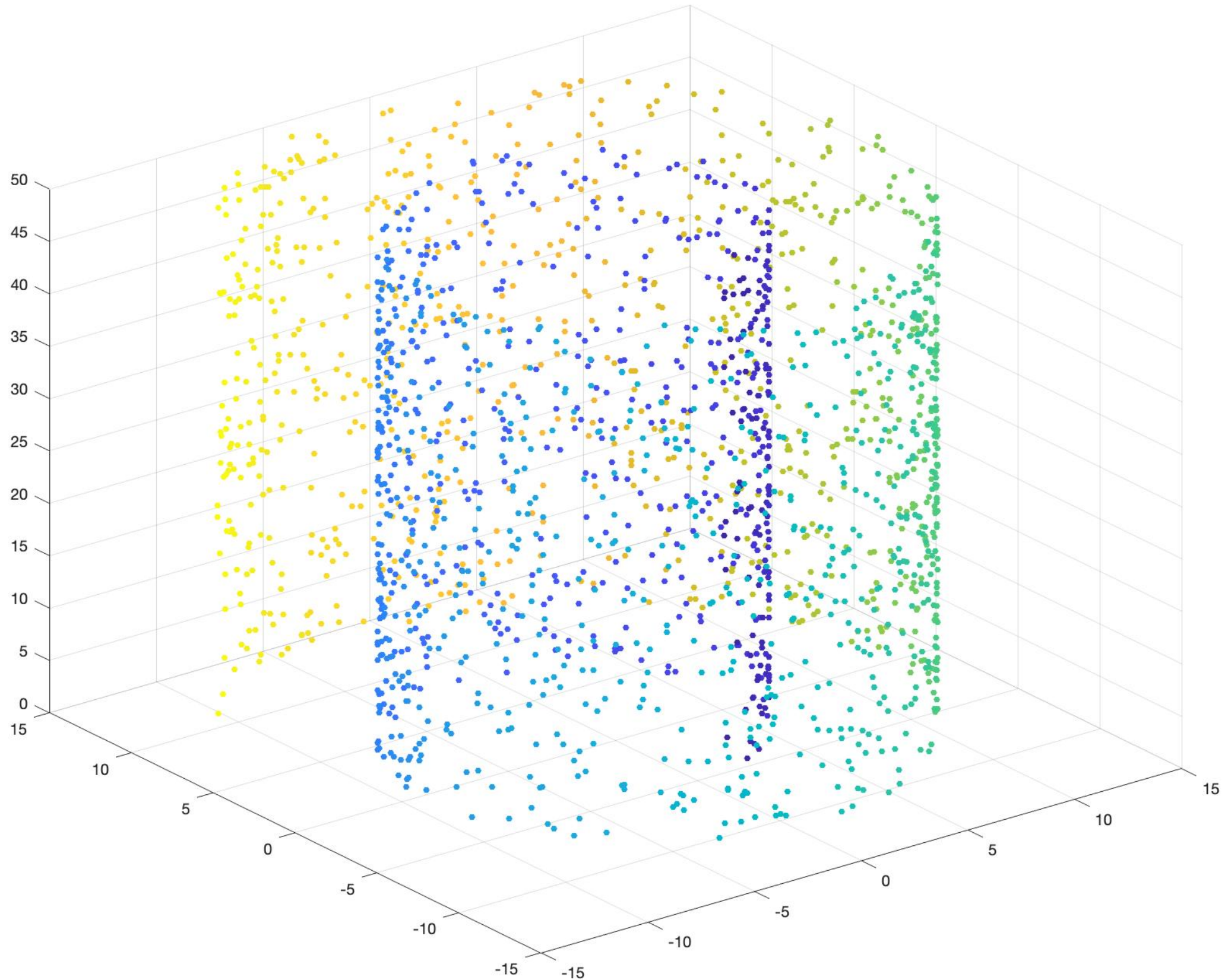
- Given m data points, $\{x^1, x^2, \dots, x^m\} \in R^n$
- Step 1: build a **weighted** graph A using nearest neighbors, and compute pairwise shortest distance matrix D
- Step 3: use a centering matrix $H = I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top$ to get
$$C = -\frac{1}{2m} H(D)^2 H$$
- Step 4: compute leading eigenvectors $w^1, w^2, \dots$ and eigenvalues $\lambda_1, \lambda_2, \dots$ of C

$$Z^\top = (w^1, w^2, \dots) \begin{pmatrix} \lambda_1^{1/2} & & \\ & \lambda_2^{1/2} & \\ & & \ddots \end{pmatrix}$$

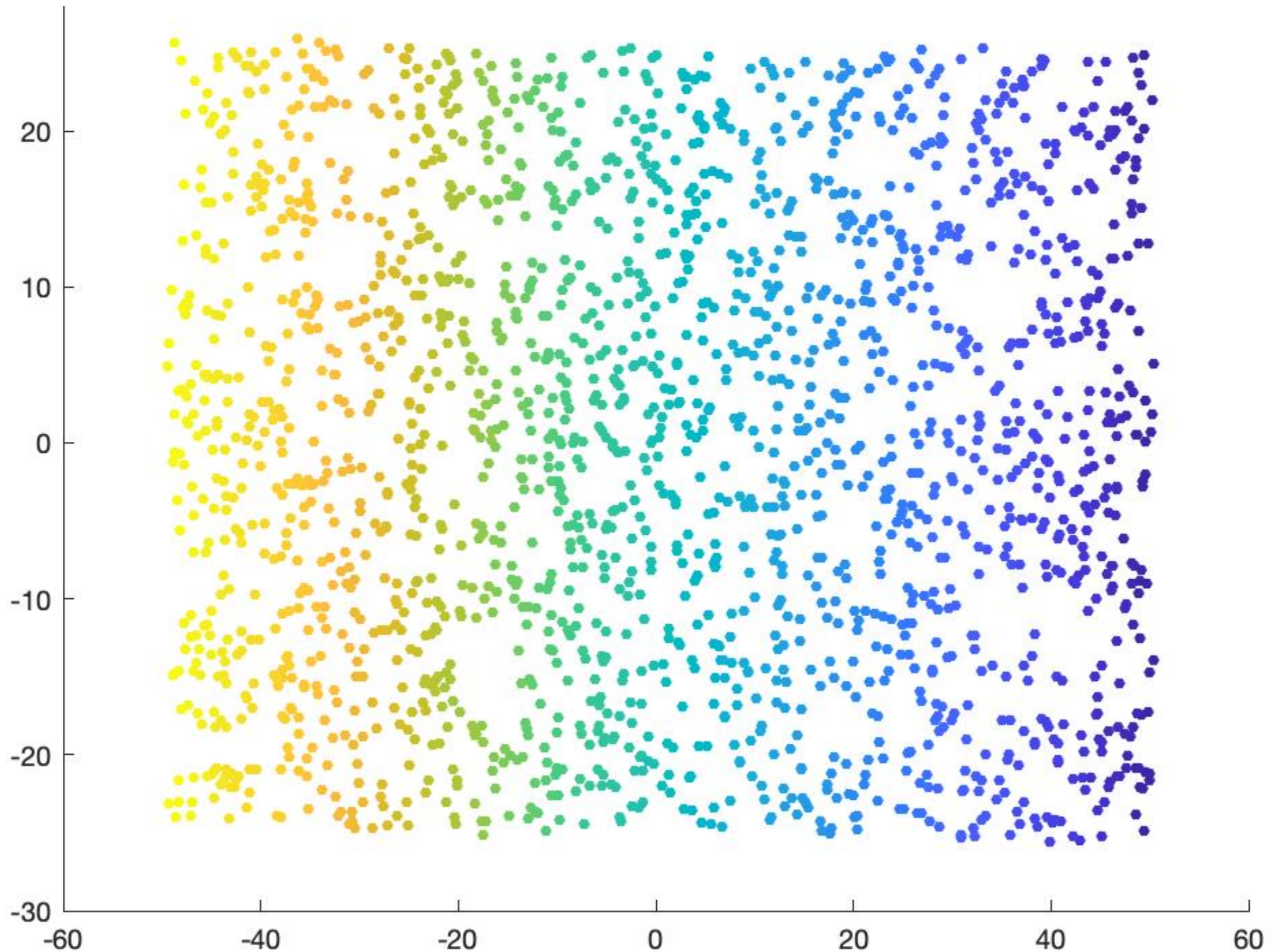
Swissroll(demo test_isomap2.m)



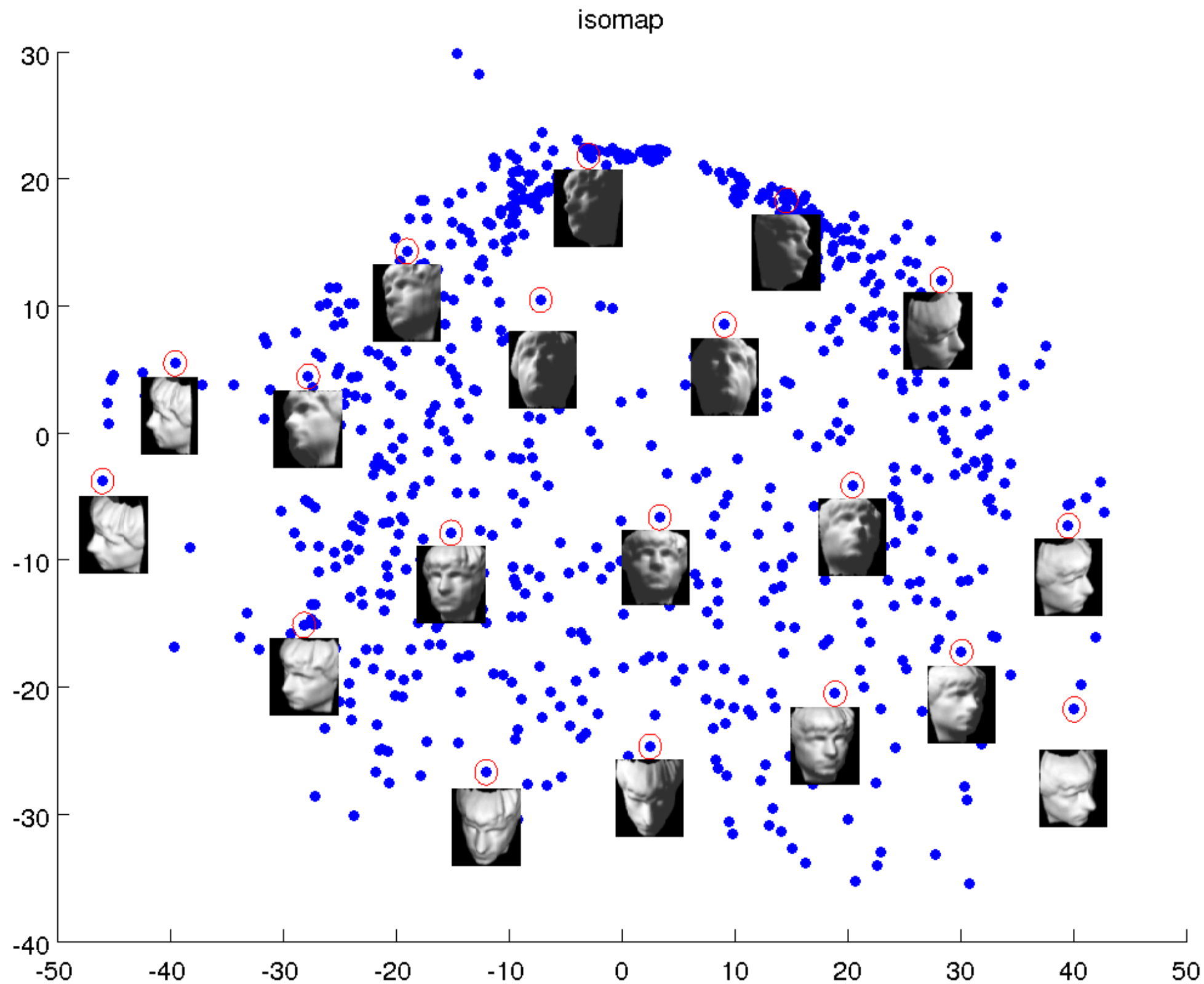
Swissroll(demo test_isomap2.m)



Swissroll(demo test_isomap2.m)



Faces (demo face_isomap_pca.m)



Handwritten digits (demo usps_isomap_pca.m)

